Angles at Centre & Circumference

Exam code: 0607

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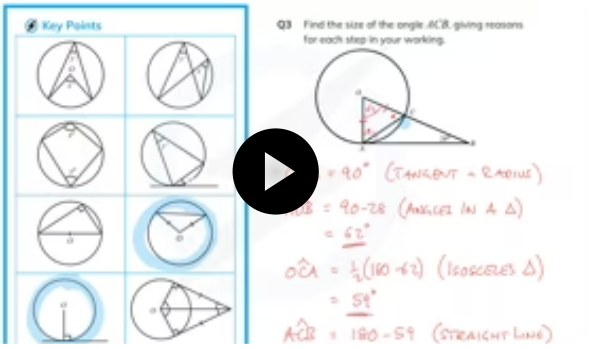

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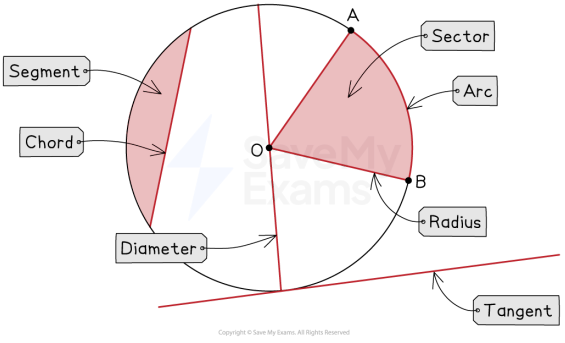
Help me understand

Angles at centre & circumference

What are circle theorems?

- **Circle Theorems** deal with angles that occur when lines are drawn within (and connected to) a circle
- You may need to use other **facts and rules** such as:
  - basic properties of lines and angles
  - properties of triangles and quadrilaterals
  - angles in parallel lines or polygons



Parts of a circle

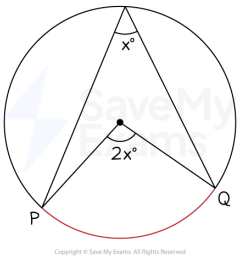
Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

circumference

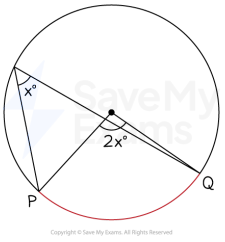
- In this theorem, the **chords (radii) to the centre** and the **chords to the circumference** are both drawn from (subtended by) the ends of the **same arc**



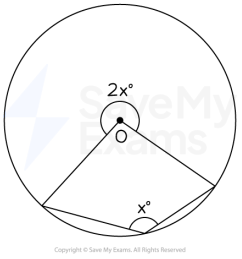
- To spot this circle theorem on a diagram
  - find any **two radii** in the circle and follow them to the **circumference**
  - see if there are **lines from those points** going to **any other point** on the **circumference**
  - it may look like the shape of an **arrowhead**

- When explaining this theorem in an exam you must use the keywords:
  - **The angle at the centre is twice the angle at the circumference**

- This theorem is still true when the **'triangle parts' overlap**



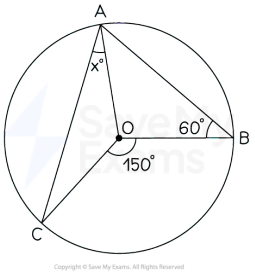
- It is also true when the lines form a **diamond shape**
  - You need to compare the **reflex angle at the centre** with the angle at the circumference
  - **Common mistakes** are to
    - compare the wrong angles
    - confuse it with a different circle theorem on cyclic quadrilaterals



Quote an **angle fact** or **circle theorem** for **every** angle you find (not just one for the final answer).

Worked Example

Find the value of x in the diagram below.



Not drawn accurately

Give a reason for each step of your working.

Answer:

There are three radii in the diagram, *AO*, *BO* and *CO*
Mark these as equal length lines

Notice how they create two isosceles triangles
Base angles in isosceles triangles are equal

Angle *OAB* = angle *OBA* = 60° (isosceles triangle)

Use the circle theorem:

The angle at the centre is twice the angle at the circumference

Form an equation for x

$$2(x + 60) = 150$$

Expand the brackets and solve the equation

$$2x + 120 = 150$$
$$2x = 30$$
$$x = 15$$

$$x = 15$$

circumference

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